Catechism of Railroad Operation

By Angus Sinclair

Questions and Answers.
Second Series.

(Continued from page 245.)

90. Under what circumstances is it proper to keep the fire door partly open when an engine is working?

A. When the fire becomes fouled by clinkers and ashes and the grates are stuck so that nothing can be done to increase the admission of air through the grates. When an unexpected call comes to shut off steam and the fire is burning vigorously.

91. Why is it necessary to carry a thicker fire in some engines than in others?

A. Engines with small nozzles and limited grate area will tear holes in a thin fire unless there is plenty of coal on the grates.

92. Is it possible to admit too much air to a fire?

A. Yes. The steam making for any train calls for a certain volume of air to be used up in combustion. If more air is used it cools the gases and tends to keep them below the igniting temperature.

93. How much air should be admitted to burn coal to the best advantage?

A. The volume of air that will make the boiler steam most freely. That may be counted on as giving the best results in the combustion of coal.

94. When the fire gets clogged up with ashes and clinkers during a long run, which is the better plan, to struggle along with dirty fire or to stop and clean the fire?

A. Much more time may be wasted struggling to make steam with a dirty fire than will be lost in stopping long enough to clean the fire.

95. Are any precautions necessary before beginning to shake the grates or to operate self-dumping ash pans?

A. Neither of these operations should be carried on when the engine is passing over bridges, cattle guards, or while crossing switches, interlocking connections, or places where fires might readily be started.

96. What is wrong when smoke trails back over the train when steam is shut off?

A. That annoying condition is due principally to heavy atmospheric conditions which prevent the smoke and gases from rising. The fireman ought to do his best to prevent the emission of smoke by following the most approved rules concerning the prevention of smoke.

CONCERNING INJECTORS AND WATER FEEDING.

97. What is an injector?

A. An apparatus employed to force water into a boiler.

98. Explain the principle upon which the injector works.

A. The action of the injector is the utilizing of the principle of induced currents to perform the work of moving water against resistance. The mechanism of the injector makes it possible to use a jet of steam to transform its high velocity to put a body of water into such high motion that it overcomes the pressure inside the boiler and keeps pouring inside the same. Two forces are represented, the pressure upon the check valve tending to resist any effort to push it away from the seat, and the water passing through the injector at high speed representing great momentum which the pressure upon the check valve is too light to resist. The scientific explanation is that static energy is overcome by kinetic force, like the quiet repose of a mill wheel being forced into motion by a rush of moving water.

99. What is the purpose of providing an injector with an overflow arrangement?

A. The overflow opening, which offers no more than atmospheric resistance, permits the water to be forced into rapid motion, thereby acquiring the momentum that enables it to force its way into the boiler.

100. What is the difference between a lifting and a non-lifting injector?

A. In a lifting injector a partial vacuum is created which raises or sucks the water from the level of the feed pipe to the level of the injector. In a non-lifting injector, the instrument is so located that the feed water will flow into its tubes by gravity.

101. What structural difference exists between a lifting and a non-lifting injector?

A. A lifting injector is provided with two tubes, one for raising the water, the other for giving it the motion which sends the water into the boiler. The non-lifting injector has not any water lifting tube.

102. What are the principal parts of an injector?

A. The principal parts of an injector are: the nozzles, which perform the functions of delivering or forcing the water into the boiler, and the operating mechanism, such as the lifting valve, steam valve, water valve and overflow valve.

103. Will an injector work with a leak between the injector and the tank?

A. A lifting injector is not likely to prime when there is such a leak.

104. Why will an injector fail to prime having a leak between the instrument and the tank?

A. Because the lifting suction tube will draw air instead of lifting water.

105. If it primes well, but breaks when steam is turned on wide, where would you expect to find the cause of the trouble?

A. In the supply of water. Would disconnect the hose and examine all its sources of water supply. The injector might prime with a limited supply of water but would break when steam was turnred on full.

106. If the injector will not prime where would you look for the cause of trouble?

A. First to the water supply. If that were found good, would look elsewhere.

107. What are the principal causes which prevent an injector from working?

A. The most common causes are: Leak in the suction pipe. Obstructed strainer or strainer too small. Coating of the nozzles by incrusting matter. Loose hose lining. Obstructions in the nozzles, such as pieces of coal or other foreign matter washed in from the tank. Obstructions in the delivery pipe, such as a sticking boiler check which will not open freely. Tank cover closed so thoroughly that air cannot enter to replace the space the water occupied. Leaky steam valve and leaky boiler, which will prevent the starting of the injector through heating the suction pipe. Hot feed water will also sometimes prevent the injector from operating.

108. Will the injector prime if the check valve leaks badly? If the injector throttle leaks badly?

A. No.

109. If steam appeared at overflow when priming valve was closed, how would you tell if the leak was from the throttle or priming valve? How would you know if the check valve was leaking?

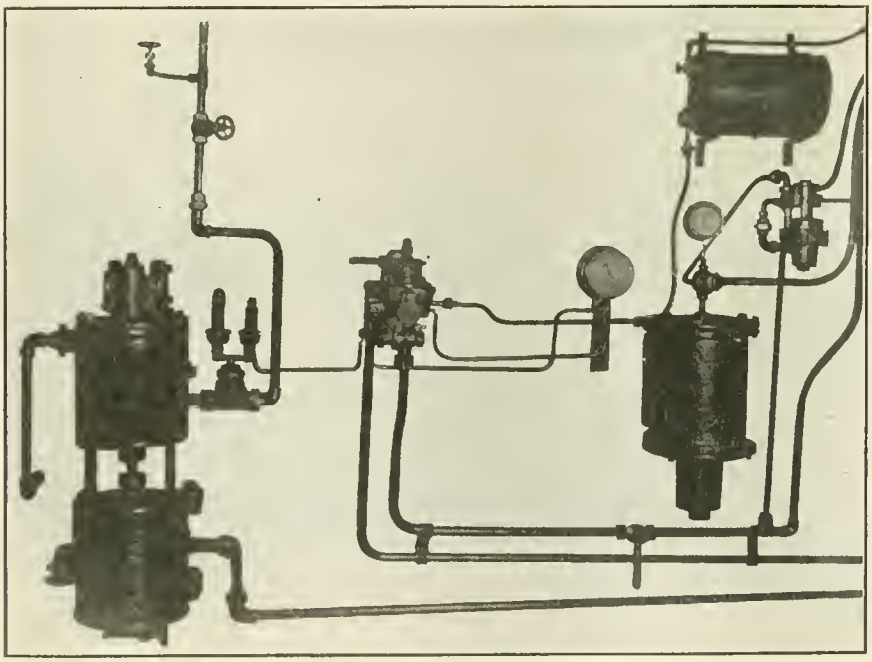
A. Opening the frost cock, with which most delivery pipes and check valve cases are provided, will give indication if the check valve is leaking. To find out if the injector steam valve is leaking, close the valve, and that will stop the leak if the throttle was at fault.

110. Will an injector prime if primer

brake pipe and bleed the pressure chamber; close pressure chamber and open the brake pipe stop cock and note the time required to charge the pressure chamber to 70 lbs. From 55 to 90 seconds time will be satisfactory.

2nd test.—Close the stop cock in the brake pipe, bleed the pressure chamber, remove the equalizing valve cylinder cap and insert the wooden washer to block the equalizing valve in lap position. Replace cover and open stop cock in brake pipe. Piston ring leakage will be shown by gauge hand attached to the pressure chamber; it should not exceed 35 or 40 lbs. per minute. On repaired work it should be kept down to 25 or 30 lbs. per minute.

3rd test.—After the wooden washer has been removed and the pressure chamber recharged, close the stop cock in the distributing valve supply pipe, place the independent brake valve handle in application position and close the stop cock in the brake cylinder brake pipe. The brake cylinder gauge hand rising will show leakage past the application piston packing leather and ring.



AIR PUMP AND BRAKE VALVE TEST RACK.

4th test.—There being no leakage past the leather, open the stop cock in the brake cylinder pipe, then slowly open the stop cock in the supply pipe, which will fill the brake cylinder; then screw the ¾-in. pipe plug in the exhaust port of the distributing valve and place the independent brake valve in running position. A blow at the emergency exhaust port of the automatic brake valve, accompanied by a fall in brake cylinder pressure after application cylinder pressure has escaped, indicates leakage past the application piston ring, which is a leak of brake cylinder pressure into the application cylinder.

Test 5.—After test No. 4, remove the plug from the distributing valve exhaust port and use the brake, making light, independent applications. The hands on the gauges showing application chamber pressure and brake cylinder pressure should rise together and show the same pressures simultaneously. In the event of a failure to do so the differences in pressure show the amount of frictional resistance encountered in moving the application piston and its attached valves.

Test 6.—Release with the independent valve and operate the distributing valve with light reductions from the automatic brake valve. The hands on the gauges showing brake pipe pressure and pressure chamber pressure should fall together, remaining at the same figure until the point of equalization is reached. Failure to do so indicates undue resistance to movement in the equalizing portion of the distributing valve. The distance the brake pipe gauge hand falls before the pressure chamber hand starts shows approximately the amount of frictional resistance.

Test 7.—Place the automatic brake valve in service position until equalizing reservoir pressure has entirely escaped. There should then be no further escape of air from the brake pipe exhaust port. If there is a continuous blow of air from the exhaust port after equalizing reservoir pressure has escaped, it indicates leakage from the brake cylinder into the brake pipe past the check valve or the guide in the quick action cylinder cap. Very slight leakage at this time accompanied by a falling of pressure chamber air as indicated by the gauge attached would point to leakage of pressure chamber air past the equalizing valve packing ring and cylinder cap gasket into the brake pipe.

Test 8.—Place the automatic brake valve in release position until brake pipe and pressure chamber have charged to main reservoir pressure, then make a light reduction in brake pipe pressure and return the handle to running position. The brake should remain applied. If the brake releases under the conditions cited, application cylinder pressure escaping at the exhaust port of the automatic brake valve, it indicates a leaky graduating valve. Graduating valve leakage, if any, might possibly be observed while the distributing valve is being operated with light reductions from the automatic brake valve, but test No. 8 will show positively whether the equalizing valve is being forced to release position due to graduating valve leakage.

Testing the application piston packing leather consists of maintaining 45 lbs in the application cylinder which is shown by the gauge connected to the application chamber, and the closed stop cock in the supply pipe prevents main reservoir pressure from entering, and any leakage past the leather and ring will quickly build up a pressure in the small space between the application piston and the closed cock in the brake cylinder pipe, and the cylinder gauge connection being between the closed cock and the distributing valve will permit the gauge to register this leakage, if there is any.

One brake pipe is used for both racks, and by placing the cut-out-cock handles in the proper positions the distributing valve can be operated with the G 6 brake valve, or, by cutting out accordingly, the triple valve can be operated with the H 6 brake valve. This is of no particular advantage except to reduce the amount of piping required, but it will be observed that if the graduating valve of a distributing valve is leaking so badly that the equalizing valve is forced to release position after an application and return to lap position, the distributing valve will release when being operated with the G 6 brake valve when that valve handle is placed on lap position after an ordinary brake pipe reduction.

The time required to bolt the distributing valve to the reservoir and conduct the tests prescribed, and remove it is about 13 minutes, provided nothing is found wrong during the test, and inasmuch as the valve cannot be properly cleaned in less than 35 or 40 minutes it follows that a 13-minute test is of a comparatively short duration. The globe valves shown in the gauge pipes have no connection with the distributing valve test, they are for the purpose

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(*Continued from page* 301.)

128. Why is the present brake called an automatic brake?

A. Because it is automatic in its action; that is, its normal condition is when it is held off, due to the maintenance of train line pressure, and anything which happens to reduce the train pipe pressure will cause the brake to apply of its own accord or automatically.

129. Where is the compressed air stored?

A. In the main reservoir on the engine, in the auxiliary reservoirs under each car, in the train line which extends throughout the train under the cars, and connects the brake valve with the triple valves connected with the auxiliary reservoirs.

130. What are the essential parts of the air brake system as applied to a locomotive?

A. The air compressor, the air compressor governor, the main air reservoirs and the engineer's valve, a duplex air pressure gauge, a plain triple valve, auxiliary reservoir, brake cylinder with its piston and the necessary piping with stop cocks and angle cocks.

131. Name the positions of the engineer's valve.

A. Release, running, lap, service application and emergency application.

132. How is the automatic brake applied and released?

A. First by putting the engineer's valve in the application position. To release the brake the handle is turned to the release position.

133. Where are the different pressures stored?

A. The highest pressure air is stored in the main reservoirs on the engine. The working pressure is in the brake pipe and in the auxiliary reservoirs under the cars.

134. What is excess pressure?

A. The amount of pressure contained in the main reservoir higher than that of the train line pressure.

135. What is the purpose of carrying excess air pressure?

A. It provides the means of releasing and recharging the brakes quickly.

136. Where do these pressures begin and end?

A. They begin with the air compressor and end with the release of brakes through the triple valves.

137. What is the use of the air compressor governor?

A. To shut off the flow of steam to the compressor when the desired main reservoir pressure has been attained.

138. What is the purpose of the main reservoirs?

A. The principal purpose of the main reservoirs is to hold a volume of air sufficient for operating the brakes on locomotives and cars. They supply the brake pipe and auxiliary reservoirs with the air required, as it tends to reduce the temperature of the air which is heated by compression of pumping.

139. Where are the main reservoirs located?

A. Generally between the frames of the engine behind the cylinder saddle or in any other convenient place.

140. What is the purpose of the auxiliary reservoir?

A. The auxiliary reservoir, which is located under each car, holds a storage of compressed air for supplying the brake cylinder with pressure to operate the piston in the brake cylinder which applies and releases the brakes.

141. What is the purpose of the brake pipe?

A.—The brake pipe is the circulating medium for air between the main reservoir and the auxiliary reservoirs. It conveys the required pressure throughout the train, and any accident that breaks it leads to the application of all the brakes.

142. What is a triple valve?

A.—The triple valve is an apparatus consisting of pistons and slide valves that apply and release the brakes, acting on the difference of pressure in train pipe and auxiliary reservoir. The triple valve performs a three-fold function. When the engineer's valve is in release position the triple valve permits air to pass from the train pipe into the auxiliary reservoir. In application position of the engineer's valve, the triple valve permits air pressure to pass from the auxiliary into the brake cylinder, thereby applying the brakes. When the engineer's valve is put into release position, the triple valve permits the air to escape from the brake cylinder to the atmosphere.

143. What is the brake cylinder and what is its use?

A.—The brake cylinder is provided with a piston which connects the brake actuating mechanism. When air is admitted to the brake cylinder through the action of the triple valve, the piston is pushed out, moving a lever that operates and applies the brake.

144. How should the air compressor be started?

A.—Very slowly with all drain pipes wide open. After the compressor is heated and the water drained off and when a pressure of 35 to 40 lbs. has accumulated in the main reservoir, the compressor throttle should be opened sufficiently to run the pump at a speed which will maintain the required pressure and perform the brake work satisfactorily. The steam end of the compressor should be lubricated freely during starting just after the drain cocks are closed.

145. What kind of oil should be used in the cylinders of an air pump?

A.—Good valve oil.

146. How often should the air end of the pump be oiled?

A.—Often enough to prevent the pump from groaning.

147. Where does the pump deliver air?

A.—To the main reservoir on the engine.

148. How can a disabled brake be cut out?

A.—By turning the stop cock on the branch or cross-over pipe.

149. How should the handle of cut-out cock stand when closed?

A.—Parallel with pipe.

150. How should handle of angle cock stand when closed?

A.—At right angles with the pipe.

151. What does the line or mark at end of plug cock indicate, regardless of position of handle?

A.—This line or mark indicates the direction of the passage way through the plug cock and by it may be known if the cock is open, regardless of the handle itself.

152.—How should a brake be bled off?

A.—The release valve in the auxiliary reservoir should be sharply opened for an instant, then quickly closed. This operation may be repeated until the triple valve begins to discharge air, which can be heard at the retaining valve or exhaust port of the triple, then no farther opening of this valve should be made.

153. What do the red and black pointers on the air gauge indicate?

A.—The red indicates main reservoir pressure; the black indicates brake pipe pressure.

Iron or steel may be made rustproof by boiling in 1 gal. of water to which is added 4 oz. of phosphoric acid and 1 oz. of iron filings. A black non-corroding coating is produced.